

Section B

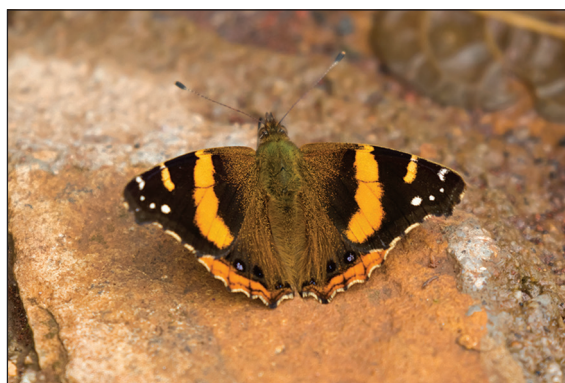
Refer to the separate Resource Folder to answer question 8.

8. (a) (i) The Red Admiral butterfly competes with greenfly for food. The scientific name for one type of greenfly is *Acyrtosiphon pisum*.

Complete the table below to show the scientific classification of greenfly. [3]

Kingdom:	
Phylum:	
Class:	Insecta
Order:	Hemiptera
Family:	Aphididae
Genus:	
Species:	
Scientific name:	<i>Acyrtosiphon pisum</i>

- (ii) Another butterfly is found in Africa. Its scientific name is *Vanessa abyssinica*.



- I. State **one** advantage of using scientific names. [1]

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- II. State **one** piece of evidence, from its scientific name, that shows it is related to the Red Admiral. [1]

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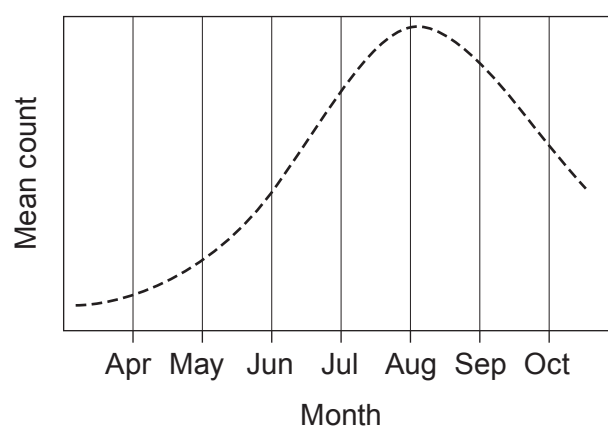
(b) The variation in population of adult Red Admirals is shown by the graph in **Figure 1**.

- (i) I. State the name of the seasonal movement of animals as seen in Red Admirals. [1]

II. State **one** reason why this seasonal movement occurs. [1]

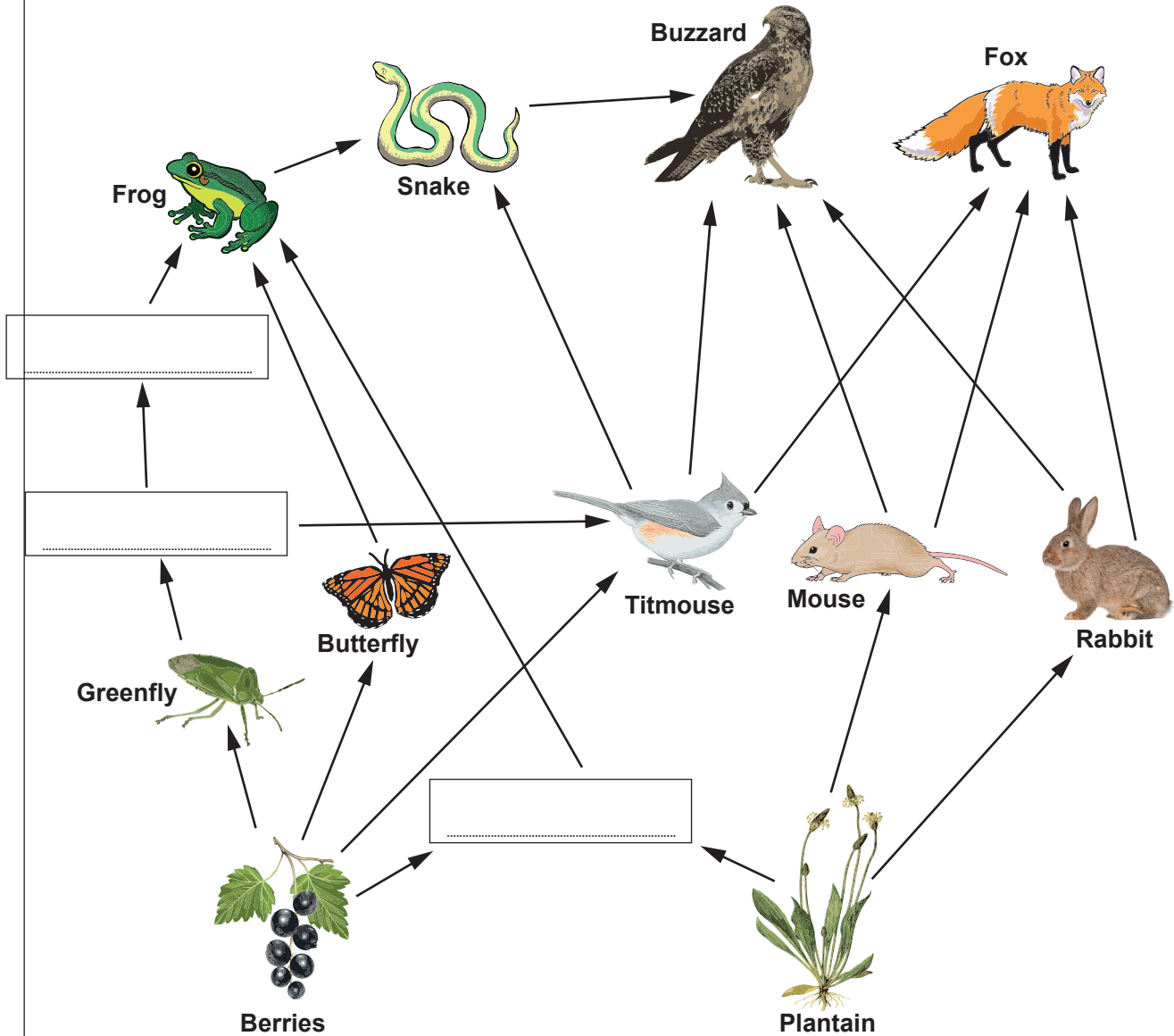
- (ii) Use the information in the graph in **Figure 1** to compare the variation of the Red Admiral population in 2019 with the mean change since 1976. [2]

- (iii) Red Admirals are a food source for frogs. **Add a line** to the graph below to show how the population of frogs changed between April and October in 2019. [2]



(c) **Table 2** gives information about the feeding relationships in one woodland habitat.

(i) Use the information in **Table 2** to **complete** the food web below. [3]



(ii) In this habitat, the fields containing plantain are cleared to make way for housing. Explain how this affects the foxes. [3]

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(iii) Explain why the frog can be found at more than one trophic level. [2]

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(d) Refer to **Table 2** to construct a pyramid of biomass for a food chain of **five** trophic levels starting with berries and ending with buzzards. [3]

(e) Use the information in **Figure 2** to calculate the efficiency of energy transfer between the primary and secondary consumers. [3]

% efficiency =

END OF PAPER



THE PERIODIC TABLE

Group

3 4 5 6 7 0

1 H Hydrogen 1

7 Li Lithium 3	9 Be Beryllium 4
23 Na Sodium 11	24 Mg Magnesium 12
39 K Potassium 19	40 Ca Calcium 20
86 Rb Rubidium 37	88 Sr Strontium 38
133 Cs Caesium 55	137 Ba Barium 56
223 Fr Francium 87	226 Ra Radium 88
	227 Ac Actinium 89

11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86

59 Co Cobalt 27	56 Fe Iron 26	55 Mn Manganese 25	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30
103 Rh Rhodium 45	101 Ru Ruthenium 44	99 Tc Technetium 43	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48
192 Ir Iridium 77	190 Os Osmium 76	186 Re Rhenium 75	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number

